

Edge AI Plant Care Monitor

Project Proposal · Arduino UNO Q (Ubuntu MPU + STM32/Zephyr MCU)

Overview

An embedded, on-device system that identifies a plant, assesses its health from a camera image, and fuses that with live soil sensors to produce a daily care recommendation. All inference runs locally on the UNO Q; no cloud dependency.

Two-Stage Concept

- 1. Recognition + Health.** A camera image is classified for species, then for health (healthy / stressed / diseased) using lightweight CNNs.
- 2. Evaluation + Recommendation.** Species and health labels are fused with soil moisture and pH, compared against per-species ideal ranges, and turned into a concrete daily action (e.g. water, hold off, adjust pH, inspect for disease).

System Architecture

- **STM32 / Zephyr (real-time):** ADC sampling of capacitive soil-moisture and pH probes at 1 Hz, calibration to engineering units, and transmission of a packed struct to the MPU.
- **Qualcomm MPU / Ubuntu (compute):** camera capture (V4L2), preprocessing to 224×224 , two TFLite inferences, sensor fusion, rule engine, and logging.
- **Bridge:** inter-processor link carrying a fixed packet `{uint32 timestamp, double moisture, double ph, uint8 status}` (21 bytes, packed).

Models & Data

- **Species:** PlantNet-300K / PlantCLEF (broad species coverage).
- **Health:** PlantVillage (54,306 images, 38 classes) for pretraining; PlantDoc for real-world robustness testing.
- **Fine-tuning:** one week of our own healthy-vs-stressed photos (one sacrificial plant stressed by withholding water, one healthy control) to close the lab-vs-classroom gap.

Target Plants

Tomato (dwarf/patio), pepper (compact), strawberry (everbearing) — all classroom-sized in 6 in pots, visually distinct for the classifier, and represented in PlantVillage for off-the-shelf health data.

Bill of Materials (core)

- Arduino UNO Q (Ubuntu MPU + STM32 MCU)
- Capacitive analog soil-moisture sensor (3.0 V output → STM32 ADC)
- pH sensor kit (BNC electrode), measuring drainage/reservoir water continuously
- MIPI-CSI carrier board (JMEDIA → 22-pin RPi pinout) + RPi-compatible camera module (IMX219 recommended for documented device-tree support)

Timeline (5 weeks)

Wk 1: hardware bring-up, MCU ↔ MPU comms, buy plants, begin daily photo/sensor logging

Wk 2: species model: assemble data, train + quantize, deploy TFLite on-device

Wk 3: health model + sensor pipeline; calibrate probes; integrate Stage 1

Wk 4: per-species reference table + rule engine; integrate Stage 2; logging

Wk 5: fine-tune on collected photos, end-to-end testing, tuning, demo prep

Risks & Mitigations

- **Camera enablement is new** on the UNO Q (device-tree edits, limited supported modules) — budget time in Wk 1; default to the documented IMX219 path.
- **Continuous soil pH is hard** — measure runoff/reservoir water, or fall back to periodic manual slurry readings if needed.
- **Lab-clean training data** underperforms on real images — mitigated by Wk 1–3 photo collection and Wk 5 fine-tuning.